INTERVIEW PREPARATION GUIDE

Data Analyst / MIS Analyst

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S E C T I O N 0 1

Python

Data Analysis & Programming Fundamentals

Data Analyst / MIS Analyst — Interview Preparation Guide

Q1. What is data analysis?

Data analysis is the process of collecting, cleaning, transforming, and interpreting data to uncover patterns, draw conclusions, and support decision-making. It converts raw data into actionable insights using tools such as Python, SQL, Excel, and Power BI.

Q2. What is Python and why is it used in data analysis?

Python is a high-level, readable programming language widely used in data analysis due to libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn. It supports data cleaning, visualisation, automation, and machine learning, making it the preferred tool for analysts.

Q3. What is a Pandas DataFrame?

A DataFrame is a two-dimensional tabular data structure with labelled rows and columns, similar to an Excel sheet. It is the core Pandas object used for data manipulation, filtering, grouping, and analysis.

Q4. How do you handle missing values in Python?

Detect missing values using df.isnull().sum(), then handle accordingly:

```
df.dropna()  — remove rows containing nulls
df.fillna(0)  — replace nulls with a constant
df['col'].fillna(df['col'].mean())  — impute with column mean
```

Q5. How do you merge two DataFrames?

```
pd.merge(df1, df2, on='key', how='inner')
```

The how parameter accepts inner, left, right, or outer — directly equivalent to SQL JOIN types.

Q6. What is a lambda function?

A lambda is a concise, anonymous function defined in a single line using the lambda keyword.

```
square = lambda x: x ** 2
```

Commonly used with df.apply() and df.map() to transform column values inline.

Q7. How do you read CSV and Excel files?

```
pd.read_csv('file.csv')
pd.read_excel('file.xlsx', sheet_name='Sheet1')
```

Both return a Pandas DataFrame ready for immediate analysis.

Q8. How do you group and aggregate data?

Use groupby() followed by an aggregate function:

```
df.groupby('Region')['Sales'].sum()
```

This is equivalent to SQL's GROUP BY clause and can be combined with agg() for multiple aggregations.

Q9. What libraries are used for data visualisation in Python?

Matplotlib handles foundational charts such as line, bar, and scatter plots. Seaborn builds on Matplotlib for statistical visuals like heatmaps, box plots, and pair plots. Plotly produces interactive charts. All three integrate seamlessly with Pandas DataFrames.

Quick Reference: Key Formulas & Commands

Import libraries

```
import pandas as pd;  import numpy as np
```

Read CSV / Excel

```
df = pd.read_csv('file.csv')   |   df =
pd.read_excel('file.xlsx')
```

Inspect data

```
df.head()  |  df.shape  |  df.info()  |  df.describe()
```

Select columns

```
df['col']   |   df[['col1', 'col2']]
```

Filter rows

```
df[df['Sales'] > 1000]
```

Handle nulls

```
df.dropna()    |    df.fillna(0)
```

Group & aggregate

```
df.groupby('Region')['Sales'].sum()
```

Merge DataFrames

```
pd.merge(df1, df2, on='ID', how='inner')
```

Apply transformation

```
df['col'].apply(lambda x: x * 2)
```

Sort values

```
df.sort_values('Sales', ascending=False)
```

Rename columns

```
df.rename(columns={'old': 'new'}, inplace=True)
```

Export to CSV

```
df.to_csv('output.csv', index=False)
```

S E C T I O N 0 2

SQL

Data Querying & Database Management

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Q1. What is SQL and what are its main components?

SQL (Structured Query Language) manages and queries relational databases. Main components: DDL (CREATE, ALTER, DROP), DML (SELECT, INSERT, UPDATE, DELETE), and DCL (GRANT, REVOKE). Data analysts primarily use DML — particularly SELECT — for data retrieval and reporting.

Q2. What is the difference between WHERE and HAVING?

```
WHERE filters rows before aggregation — cannot be used
with aggregate functions.
HAVING filters groups after aggregation and must follow
GROUP BY.
```

Example: SELECT Dept, COUNT(*) FROM Emp WHERE Salary > 30000 GROUP BY Dept HAVING COUNT(*) > 5

Q3. Explain the types of JOINs in SQL.

```
INNER JOIN — returns only matching rows from both tables
LEFT JOIN  — all rows from the left table; NULLs for
unmatched right rows
RIGHT JOIN — all rows from the right table; NULLs for
unmatched left rows
FULL OUTER JOIN — all rows from both tables, NULLs where
no match exists
```

Q4. What are aggregate functions in SQL?

Functions that compute a single result from a set of rows: COUNT(), SUM(), AVG(), MAX(), MIN().

```
SELECT Region, SUM(Revenue) FROM Sales GROUP BY Region
```

Q5. What is a subquery? Give an example.

A subquery is a query nested inside another. It can appear in SELECT, FROM, or WHERE clauses.

```
SELECT Name FROM Emp WHERE Salary > (SELECT AVG(Salary)
FROM Emp)
```

Q6. What is the difference between UNION and UNION ALL?

```
UNION — combines two result sets and removes duplicate rows.
UNION ALL — combines result sets and retains duplicates (faster).
```

Both require matching column counts and compatible data types.

Q7. What are window functions in SQL?

Window functions compute values across related rows without collapsing the result set.

```
ROW_NUMBER(), RANK(), DENSE_RANK(), LAG(), LEAD(), SUM() OVER()
SELECT Name, SUM(Sales) OVER(PARTITION BY Region) FROM Sales
```

Q8. How do you find duplicate records in a table?

```
SELECT Email, COUNT(*) FROM Customers GROUP BY Email
HAVING COUNT(*) > 1
```

This is a standard data quality check before building any MIS report.

Q9. What is a stored procedure?

A stored procedure is a precompiled, reusable set of SQL statements saved in the database. It accepts parameters, returns results, and improves performance, security, and code reusability by limiting direct table access.

Q10. How do you optimise a slow SQL query?

```
Add indexes on frequently filtered or joined columns
Select only required columns — avoid SELECT *
Replace correlated subqueries with JOINs where possible
Apply WHERE filters before joining large tables
Use execution plans to identify bottlenecks
```

S E C T I O N 0 3

Excel

Reporting, Formulas & MIS Dashboards

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Q1. What is MIS reporting?

MIS (Management Information System) reporting is the structured process of collecting, processing, and presenting operational data to help management make informed decisions. It typically covers daily, weekly, and monthly reports on sales, inventory, financials, or workforce metrics. MIS Analysts maintain report templates, ensure data accuracy, and deliver outputs to stakeholders on schedule.

Q2. What are the most important Excel functions for a data analyst?

The essential functions include VLOOKUP/XLOOKUP and INDEX-MATCH for lookups; SUMIF/SUMIFS and COUNTIF/COUNTIFS for conditional aggregation; IF/IFS for logic; IFERROR for error handling; TEXT and CONCATENATE for formatting; and array functions such as FILTER, UNIQUE, and SORT available in Excel 365.

Q3. What is VLOOKUP and what are its limitations?

VLOOKUP retrieves a value from a table by matching a lookup key in the first column.

```
=VLOOKUP(A2, $D$2:$F$100, 2, FALSE)
```

Limitations: searches rightward only, sensitive to column reordering, returns only the first match. XLOOKUP or INDEX-MATCH is preferred for flexibility.

Q4. What is a PivotTable and how is it used in MIS reporting?

A PivotTable aggregates and summarises large datasets through a drag-and-drop interface — no formulas required. In MIS reporting it generates monthly performance summaries, department-wise breakdowns, variance analyses, and trend reports quickly and interactively.

Q5. How does SUMIFS work?

SUMIFS sums values that satisfy multiple criteria simultaneously.

```
=SUMIFS(D2:D100, B2:B100, "South", C2:C100, "Q1")
```

Returns total sales from the South region in Q1. More flexible than SUMIF, which supports only one condition.

Q6. What is conditional formatting?

Conditional formatting applies visual cues — colour fills, icons, or data bars — to cells based on defined rules. It is used to flag values below target, highlight duplicates, display heat maps, and draw attention to exceptions in dashboards without altering underlying data.

Q7. How do you remove duplicates in Excel?

```
Data tab → Remove Duplicates — select columns to check
and remove
Flag duplicates first:  =IF(COUNTIF($A$2:A2, A2)>1,
"Duplicate", "Unique")
Excel 365:  =UNIQUE(A2:A100)  returns a deduplicated
list dynamically
```

Q8. What is Power Query in Excel?

Power Query is Excel's built-in ETL tool. It connects to multiple data sources, cleans and reshapes data through a step-by-step graphical interface, and loads the result into the data model. All transformation steps are saved and automatically reapplied on each refresh, making it ideal for repeatable data preparation.

Q9. What is the difference between absolute and relative references?

```
Relative: A1 — adjusts automatically when the formula is
copied
Absolute: $A$1 — remains fixed regardless of where the
formula is copied
Mixed: $A1 or A$1 — locks either the column or the row
only
```

Absolute references are essential for fixed inputs such as targets, tax rates, and lookup tables.

Q10. How do you build a dynamic dashboard in Excel?

Connect PivotTables to the data source, build PivotCharts, add slicers for interactive filtering, and use timelines for date navigation. Apply consistent chart formatting and named ranges for clarity. The dashboard refreshes automatically whenever the underlying data source is updated.

Quick Reference: Key Formulas & Commands

VLOOKUP

```
=VLOOKUP(A2, $D$2:$F$100, 2, FALSE)
```

XLOOKUP

```
=XLOOKUP(A2, D2:D100, E2:E100, "Not Found")
```

INDEX-MATCH

```
=INDEX(E2:E100, MATCH(A2, D2:D100, 0))
```

SUMIF

```
=SUMIF(B2:B100, "North", D2:D100)
```

SUMIFS

```
=SUMIFS(D2:D100, B2:B100, "South", C2:C100, "Q1")
```

COUNTIF

```
=COUNTIF(A2:A100, "Completed")
```

COUNTIFS

```
=COUNTIFS(B2:B100, "Sales", C2:C100, ">50000")
```

IF

```
=IF(D2>=100000, "Target Met", "Below Target")
```

IFERROR

```
=IFERROR(VLOOKUP(A2, $D$2:$F$100, 2, FALSE), "Not
Found")
```

TEXT

```
=TEXT(A2, "DD-MMM-YYYY")
```

CONCATENATE

```
=CONCATENATE(A2, " ", B2)    or    =A2&" "&B2
```

UNIQUE (365)

```
=UNIQUE(A2:A100)
```

FILTER (365)

```
=FILTER(A2:C100, B2:B100="North")
```

Duplicate flag

```
=IF(COUNTIF($A$2:A2, A2)>1, "Duplicate", "Unique")
```

S E C T I O N 0 4

Power BI

Data Modelling, DAX & Interactive Reporting

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Q1. What is Power BI and what are its main components?

Power BI is Microsoft's business intelligence platform for connecting to data, building models, and creating interactive reports. Core components: Power BI Desktop (authoring), Power BI Service (cloud sharing), Power Query (data transformation), and DAX (calculated fields and measures).

Q2. What is data modelling in Power BI?

Data modelling defines how tables relate to one another through primary and foreign key relationships. A well-designed model — typically a Star Schema with a central fact table and surrounding dimension tables — ensures DAX measures calculate correctly, visuals filter consistently, and report performance stays efficient.

Q3. What is DAX and can you give an example?

DAX (Data Analysis Expressions) is Power BI's formula language for creating calculated columns and measures.

```
Total Sales = SUM(Sales[Amount])
Profit Margin = DIVIDE(SUM(Sales[Profit]),
SUM(Sales[Revenue]), 0)
```

Measures are dynamic — they recalculate in real time based on the current filter context in the report.

Q4. What is the difference between a calculated column and a measure?

```
Calculated column — evaluated row by row during data
refresh; stored in the model; consumes memory.
Measure — evaluated dynamically at query time based on
filter context; uses no storage.
```

Use measures for aggregations (SUM, AVERAGE, COUNT); use calculated columns for values needed for slicing or filtering.

Q5. What are relationships in Power BI and why do they matter?

Relationships link tables via shared key columns and enable cross-table filtering in reports. Without correctly configured relationships, DAX measures and visuals will not filter as expected, producing inaccurate results. Power BI supports one-to-many, many-to-one, and many-to-many cardinalities.

Q6. What does Power Query do in Power BI?

Power Query extracts data from source systems, cleans it by handling nulls, renaming columns, and correcting data types, then reshapes it through merging, appending, and unpivoting. Every step is recorded as a reusable pipeline that runs automatically on each scheduled refresh.

Q7. What is the difference between a filter and a slicer?

```
Slicer — a visible interactive visual on the canvas that
users click to filter report data.
Filter — configured in the Filters pane; can be set at
visual, page, or report level.
```

Slicers are user-facing and intuitive; filters are typically used by report authors to apply fixed or hidden restrictions.

Q8. What is the CALCULATE function in DAX?

CALCULATE evaluates an expression within a modified filter context — it is one of the most powerful DAX functions.

```
Sales_2023 = CALCULATE(SUM(Sales[Amount]), Sales[Year] =
2023)
```

It is used extensively for time intelligence calculations, scenario comparisons, and conditional aggregations.

Q9. What is Row-Level Security (RLS) in Power BI?

RLS restricts data access at the row level based on the identity of the logged-in user.

```
Define a role with a DAX filter rule, e.g.:  [Region] =
USERNAME()
```

After publishing, assign users to roles in Power BI Service so each person sees only their permitted data.

Q10. How does data refresh work in Power BI?

Power BI Pro supports up to 8 scheduled refreshes per day; Premium supports 48. On-premises sources require the On-premises Data Gateway. DirectQuery queries the source live with no import. Import mode is faster but requires periodic refresh to reflect the latest data.

Q11. How do you create a KPI or summary card visual?

Drag a measure into a Card visual to display a single headline metric such as total revenue or order count. A KPI visual shows the measure against a defined target with a directional indicator. Both are standard elements in executive dashboards for at-a-glance performance monitoring.

S E C T I O N 0 5

HR Questions

Behavioural & Situational Interview Answers

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Q1. Tell me about yourself.

(Personalise this answer using the Present–Past–Future structure: your current academic or professional background, relevant projects and technical skills, and what you are looking forward to in this role. Keep it to approximately 90 seconds and focus on what is relevant to the job.)

Q2. Why do you want to work as a Data / MIS Analyst?

I am drawn to this role because I enjoy working with data to identify patterns and support informed decision-making. During my academic years I worked with Excel, SQL, and Power BI on real-world datasets and found it genuinely rewarding to convert raw numbers into meaningful insights. An analyst role allows me to do this professionally and continuously develop my skills. I believe my technical foundation and analytical mindset align well with the requirements of this position.

Q3. What are your strengths?

My key strengths are attention to detail — which is essential for maintaining data accuracy — strong analytical thinking, and the ability to learn new tools quickly. I am proficient in Excel, SQL, and Power BI, and I am comfortable handling large datasets in a methodical, structured manner. I also communicate findings clearly, which is important when presenting reports to non-technical stakeholders.

Q4. What is your greatest weakness?

I tend to review my work extensively before sharing it, which can occasionally take more time than necessary. I have been addressing this by setting internal time limits for each task and using structured checklists to review outputs efficiently without repeating work from scratch. This approach has helped me balance thoroughness with timely delivery.

Q5. Where do you see yourself in five years?

In five years, I see myself having grown into a Senior Analyst or BI Specialist role, managing end-to-end reporting pipelines and contributing meaningfully to data strategy. I aim to deepen my expertise in Power BI, advanced SQL, and Python within this organisation, and gradually take on broader, cross-functional reporting responsibilities.

Q6. Why should we hire you?

I bring a solid foundation in the tools this role requires — SQL, Excel, and Power BI — combined with genuine enthusiasm for data and reporting. I am a quick learner, detail-oriented, and comfortable working independently with large datasets. I am looking for a role where I can contribute from day one and grow steadily with the team, and I believe this position offers exactly that opportunity.

Q7. How do you handle pressure and tight deadlines?

I prioritise tasks by urgency and impact, break deliverables into manageable checkpoints, and communicate proactively if a deadline is at risk. During my final year project I compiled and analysed data from five departments within three days by creating a clear daily plan and focusing on critical outputs first — submitting on time with positive feedback.

Q8. Tell me about a time you worked in a team.

During an academic case study, I worked in a group of four analysing sales data. I handled data cleaning and dashboard creation while colleagues managed data collection and the presentation. When the team disagreed on methodology, we discussed options and reached a consensus. The project was completed successfully and received strong faculty feedback.

Q9. Are you comfortable working with large datasets and repetitive tasks?

Yes, I understand that data analyst roles involve repetitive but critical tasks such as data cleaning, validation, and scheduled report generation. I approach such work systematically — building reusable templates and queries to ensure consistency and efficiency without compromising accuracy. I consider these tasks the foundation of reliable, trustworthy reporting.

Q10. What do you know about MIS reporting?

MIS reporting involves collecting operational data, processing it, and presenting it in a structured format to support management decision-making. It typically covers daily, weekly, and monthly reports on sales, inventory, financials, and workforce metrics. MIS Analysts maintain reporting templates,

ensure data integrity, and coordinate with departments to gather inputs and deliver outputs on schedule.

Q11. How do you ensure accuracy in your work?

I cross-verify report totals against source data, apply data validation rules, and use IFERROR checks in Excel to catch formula errors. For SQL outputs, I run COUNT and SUM checks to confirm record counts match the source before finalising any report. I also conduct a structured final review against expected values prior to submission to ensure nothing is missed.

Q12. How do you manage multiple tasks simultaneously?

I maintain a prioritised task list categorised by urgency and deadline. For recurring reports I build reusable templates and automate repetitive steps to save time. When multiple deadlines converge, I communicate with stakeholders early to manage expectations and ensure nothing slips.

Q13. What are your salary expectations?

Based on my research into industry standards for this role and location, I am looking for a package in the range of [X to Y]. That said, I am open to discussion based on the overall compensation structure, growth trajectory, and learning opportunities the organisation offers. My immediate priority is finding the right environment to build a strong career in data analytics.

Q14. Do you have any questions for us?

Yes, I do have a few questions for you:

- What does a typical reporting cycle look like for the MIS team?
- Which tools and technologies does the team primarily use day-to-day?
- What does success look like in this role within the first three to six months?
- Could you please share how the onboarding and training process is structured for this role?
- What kind of growth opportunities or career path can I expect in this role over time?

Q15. Tell me about a project where you used data to solve a problem.

In my final year, I analysed three years of retail sales data for an academic case study. I used Python to clean the dataset, SQL to extract monthly regional trends, and Excel PivotTables to build a management summary dashboard. The analysis revealed a consistent 18% sales decline in Q3 attributable to seasonal demand shifts, and I recommended a targeted promotional strategy for that period. The project was well received by my faculty panel and gave me an end-to-end view of the analytics workflow.

Best of Luck with Your Interview!

Prepare thoroughly, practise your answers aloud, and walk in with confidence.

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Connect with the Author

GitHub: https://github.com/SahilS19
LinkedIn: https://www.linkedin.com/in/sarthak-raj08/

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